

# Siamese Neural Networks for One-Shot Face Recognition

## 1 Introduction

One-shot learning refers to the problem of recognizing previously unseen classes given only a single labeled example per class. This setting is particularly challenging for conventional supervised learning approaches, which typically rely on large labeled datasets to achieve robust generalization.

A seminal solution to this problem was proposed by Koch, Zemel, and Salakhutdinov (2015), who introduced Siamese Neural Networks trained on a verification task rather than direct classification. Their approach learns a similarity function between pairs of images, enabling generalization to entirely new classes without retraining. Using a convolutional Siamese architecture with a weighted  $L_1$  distance and sigmoid output, they achieved near human-level accuracy on the Omniglot one-shot classification benchmark.

While the original work focused on handwritten character recognition, extending the same methodology to face recognition introduces additional difficulties. Face images exhibit high intra-class variability due to pose, illumination, expression, and background clutter. As a result, models trained strictly according to the original paper may struggle to generalize effectively in this domain.

The goal of this project is to reproduce the original Siamese network as a baseline and then explore targeted improvements aimed at better adapting the model to face recognition. These improvements focus on input preprocessing while preserving the original one-shot learning framework.

## 2 Dataset

The dataset is derived from a face image collection organized by identity, where each identity may contain between one and several dozen images. However, the training process does not operate directly on all identity images. Instead, it relies exclusively on predefined verification pairs supplied in text files, following the standard LFW evaluation protocol. All images have a fixed spatial resolution of  $250 \times 250$  pixels and are already grayscale.

The identities in the dataset contain a minimum of one image and a maximum of 530 images, with an average of approximately 2.3 images per identity. In total, the dataset consists of 13,233 images spanning 5,749 identities.

The dataset is divided using two official pair definition files:

- `pairsDevTrain.txt`, which defines training pairs
- `pairsDevTest.txt`, which defines test pairs

Each file specifies two types of verification pairs:

- **Positive pairs:** two image indices belonging to the same identity
- **Negative pairs:** image indices from two different identities

These files explicitly control which images participate in training and testing, meaning that:

- The dataset split is pair-based rather than identity-exhaustive
- Some identities may appear with only a small subset of their available images
- There is no independent validation identity pool; therefore, a validation set is explicitly constructed from the training pairs

In the training dataset, there are 3,443 unique images corresponding to 2,132 unique identities, with a minimum of one image, a mean of 2.06 images, and a maximum of 10 images per identity. In the test dataset, there are 1,549 unique images corresponding to 963 unique identities, with a minimum of one image, a mean of 2.08 images, and a maximum of 10 images per identity. There is no overlap between training and test identities or images.

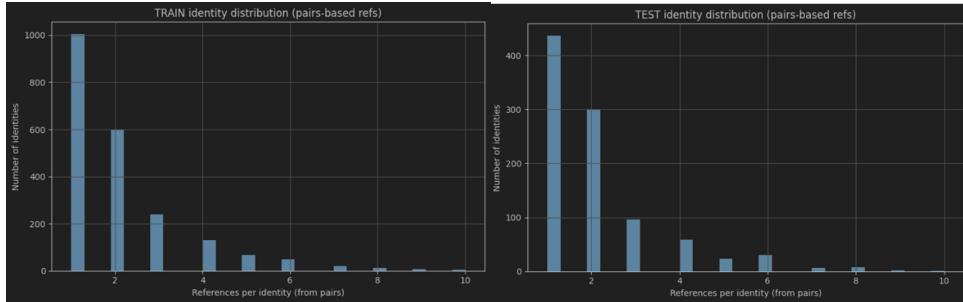


Figure 1: Identity Distribution in Train and Test.

Both the training and test pair files are constructed to be approximately balanced:

- Each file begins with a number indicating the total number of pairs
- Pairs are evenly divided between positive (same identity) and negative (different identity) examples

As a result, class balance is fixed and externally imposed, rather than inferred from the raw dataset distribution.

Exploratory Data Analysis (EDA) supports the claim that faces are tightly aligned and consistently centered within the image frame:

- Mean and variance images exhibit broad spatial activation
- Facial structure is clearly visible and sharply localized
- Background regions exhibit negligible variance

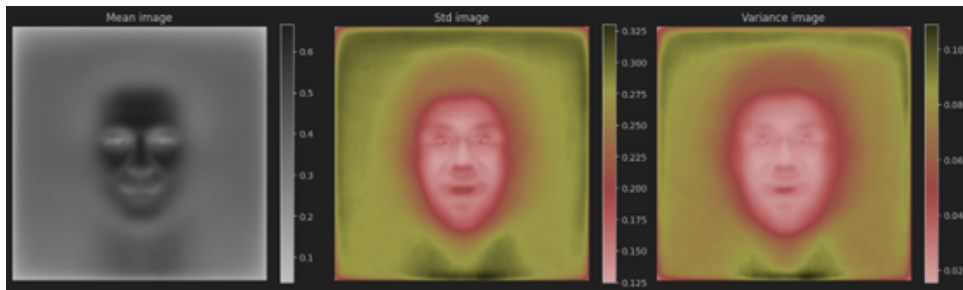


Figure 2: Mean, standard deviation, and variance images of the dataset.

This indicates strong alignment typical of LFW-style datasets, rather than studio-quality or landmark-aligned face crops. Variance is largely confined to facial landmarks such as the eyes, mouth, and nose:

- Hair, shoulders, and background regions do not contribute significantly
- Illumination variation affects only small portions of the image

Thus, the dataset contains substantial nuisance variation while remaining dominated by facial structure rather than background information.

### 3 Preprocessing

#### 3.1 Baseline Preprocessing

In the baseline experiment, preprocessing follows the original Siamese network formulation. Input images are resized to a fixed spatial resolution of  $105 \times 105$  pixels, converted to grayscale, and normalized using fixed mean and standard deviation values. The resulting tensors are then fed directly into the Siamese network without any explicit semantic filtering, localization, or background suppression.

This setup introduces minimal domain-specific assumptions and serves as a faithful reproduction of the original methodology. In the absence of explicit cropping or masking, the network is required to implicitly learn to suppress background information through supervision alone.

#### 3.2 Augmentation-Based Preprocessing

The second experiment extends the baseline preprocessing by introducing stochastic data augmentation during training, including random resized cropping, small affine transformations, horizontal flipping, grayscale intensity jitter, and occasional Gaussian blur. These additions increase robustness to geometric and photometric variation without modifying the underlying one-shot learning formulation.

All images are resized to  $105 \times 105$  pixels and converted to grayscale. Pixel intensities are normalized using fixed mean and standard deviation values to stabilize optimization.

During training, stochastic augmentations are applied to encourage invariance to minor geometric and photometric variations. These include random resized cropping with near-square aspect ratios, small affine transformations (rotation, translation, and scale), random horizontal flipping, mild grayscale intensity jitter (brightness, contrast, and gamma), and occasional Gaussian blurring. Evaluation preprocessing remains fully deterministic.

This strategy does not alter labels, loss functions, network architecture, or the one-shot learning protocol. Instead, it improves generalization by exposing the encoder to realistic intra-class variation.



Figure 3: Preprocessing pipelines side by side with original.

## 4 Experiments

### 4.1 Baseline Experiment

The baseline experiment reproduces the original Siamese network configuration as closely as possible. Images are resized, normalized, and passed directly to the network without augmentation or domain-specific preprocessing.

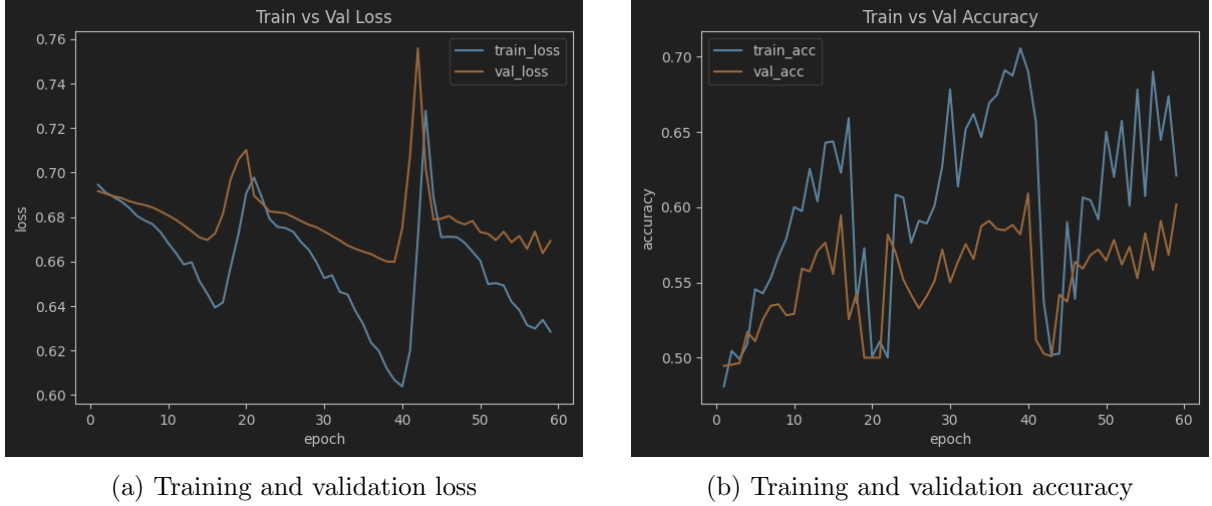


Figure 4: Learning curves of the original pipeline.

Training accuracy increases steadily but exhibits instability, with abrupt drops and spikes in both training and validation curves. Validation accuracy improves modestly and remains consistently below training accuracy, indicating limited generalization. These oscillations suggest sensitivity to optimization dynamics and increased difficulty in suppressing background variability without explicit inductive bias. The training loss reaches its minimum of approximately 0.60 around epoch 40, while validation loss stabilizes at a higher level near 0.66, indicating a persistent generalization gap. Training accuracy peaks at roughly 0.71, where as validation accuracy remains lower and plateaus around 0.60, suggesting that improvements beyond this point primarily reflect overfitting rather than better generalization.

### 4.2 Experiment with Augmentation-Based Preprocessing and face mask

The second experiment introduces the improved preprocessing pipeline that is using face mask to crop the area of the face and remove the noisy background while keeping the face intact. In Addition, I used stochastic training-time augmentations. These include random resized cropping, small affine transformations, horizontal flipping, grayscale intensity jitter, and occasional Gaussian blurring, while evaluation preprocessing remains deterministic.

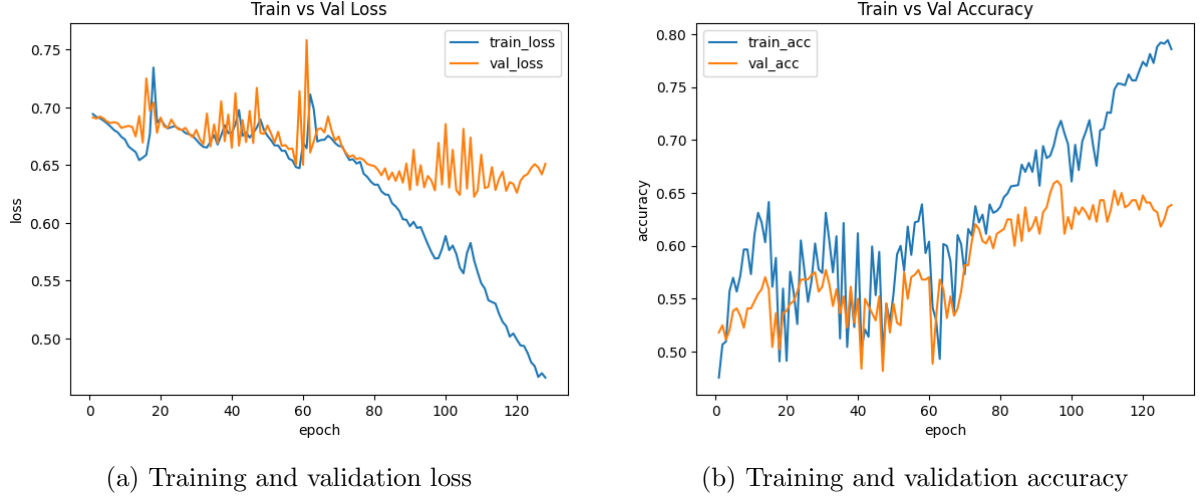


Figure 5: Learning curves of training using face masks.

Training accuracy increases more rapidly and reaches approximately 0.79, while validation accuracy plateaus around 0.63–0.65. The widening gap after mid-training indicates faster fitting followed by overfitting rather than uniform improvement. Training loss decreases substantially, while validation loss stabilizes with noticeable oscillations, consistent with a small validation set and pair-based training.

These results suggest that the preprocessing improves robustness and feature learning for the training distribution, but additional regularization or early stopping is required to further improve generalization.

## 5 False Analysis

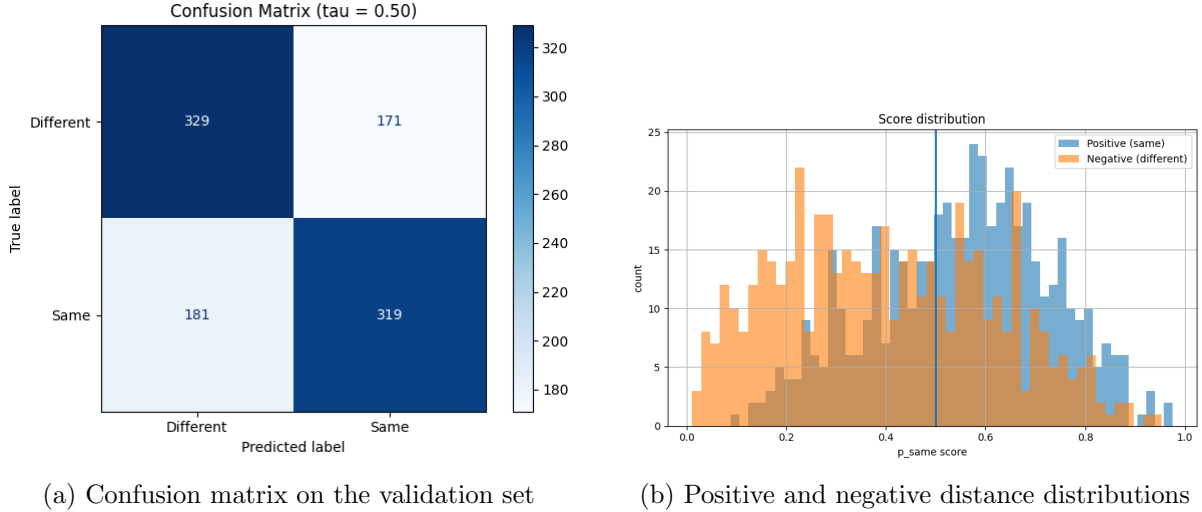


Figure 6: Evaluation of the face verification model. The confusion matrix and embedding distance distributions provide insight into classification behavior and representation quality.

**Key Observations** Figure 6 highlights several important characteristics of the model’s performance:

- The number of false positives and false negatives is approximately balanced (FP = 171, FN = 181), indicating that the classifier is not biased toward predicting either “same” or

“different” identity pairs.

- The overall verification accuracy is approximately 64.8%, which is:
  - Significantly above random chance.
  - Substantially below the expected performance of a well-trained Siamese verification model on this dataset.

**Interpretation** The balanced error rates, together with the strong overlap observed between the positive and negative distance distributions (Figure 6b), suggest that the primary limitation lies in the quality of the learned embedding space. Specifically, the model fails to establish a clear margin between same-identity and different-identity pairs. This behavior points to a *representation weakness* rather than an improperly chosen decision threshold.

## 5.1 Improved Model And Preprocessing

The third experiment extends the baseline preprocessing with stochastic training-time augmentations, including random resized cropping, small affine transformations, horizontal flipping, grayscale intensity jitter, and occasional Gaussian blurring, while evaluation preprocessing remains fully deterministic and does not employ face masking. In addition, the encoder architecture is modified by replacing ReLU activations with Leaky ReLU and by introducing batch normalization layers. These changes are intended to improve optimization stability and gradient flow, particularly in deeper layers of the embedding network, and to reduce sensitivity to activation sparsity and internal covariate shift, thereby supporting more stable training in the presence of increased input variability.

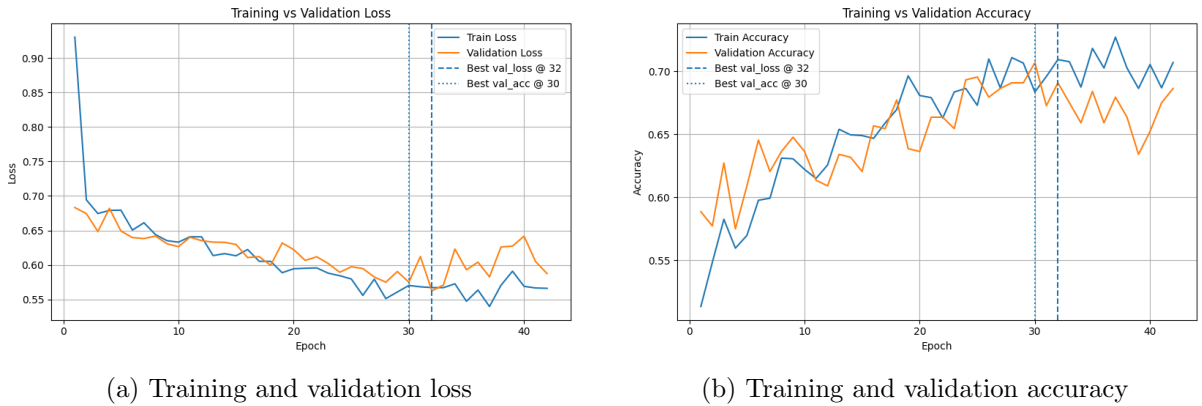


Figure 7: Learning curves of training using face masks.

Training and validation accuracy increase steadily during the early stages of training, indicating improved learning dynamics under the augmented preprocessing pipeline. Training accuracy remains consistently higher and stabilizes around 0.70–0.73, while validation accuracy reaches a plateau near 0.69–0.70 around the mid-training phase. Beyond this point, validation accuracy shows limited further improvement, whereas training accuracy continues to fluctuate at a higher level, suggesting the onset of overfitting.

The loss curves exhibit a corresponding trend. Training loss decreases steadily throughout training, reaching values close to 0.55, while validation loss decreases more gradually and attains its minimum around the same mid-training region. Afterward, validation loss displays mild oscillations without sustained improvement. Overall, these results suggest that the augmented preprocessing improves optimization and early generalization, with optimal performance achieved near the validation minimum, motivating the use of early stopping.

## 6 Takeaways

This project investigated the application of Siamese Neural Networks to one-shot face recognition, with a particular focus on the impact of preprocessing and architectural choices. While the original Siamese formulation provides a strong conceptual framework for verification-based learning, the experiments demonstrate that its performance on face data is highly sensitive to both input representation and optimization stability.

Augmentation-based preprocessing was shown to improve training dynamics and early generalization by exposing the model to realistic geometric and photometric variability. However, explicit face masking was found to be detrimental, likely due to the removal of contextual cues and the introduction of artifacts that interfere with similarity learning. These findings highlight that, in verification settings, aggressive spatial priors may reduce robustness rather than improve it.

A key limitation identified in this work is the simplicity of the encoder architecture inherited from the original paper. The shallow convolutional embedding network, while sufficient for Omniglot-style character recognition, appears underpowered for face recognition, which involves significantly higher intra-class variation. The observed performance ceilings and early saturation of validation accuracy suggest that representation capacity, rather than supervision or data quantity, is a primary bottleneck.

Future work should therefore prioritize replacing the encoder with a more expressive architecture, such as a deeper convolutional network or a lightweight residual model, while preserving the Siamese verification framework. Incorporating modern architectural components—such as residual connections, deeper normalization schemes, or larger embedding dimensions—may allow the model to better capture identity-relevant structure without sacrificing generalization.

Additional preprocessing improvements may also be explored, including stronger but carefully controlled augmentations, or curriculum-based augmentation schedules. Finally, regularization techniques such as weight decay tuning, earlier stopping based on validation behavior, or margin-based losses could further improve generalization. Together, these directions suggest that while the Siamese framework remains well-suited for one-shot learning, effective face recognition requires both stronger representations and preprocessing strategies tailored to the complexity of the domain.

## References

- [1] Koch, G., Zemel, R., & Salakhutdinov, R. (2015). Siamese Neural Networks for One-Shot Image Recognition.